

# Machine Learning

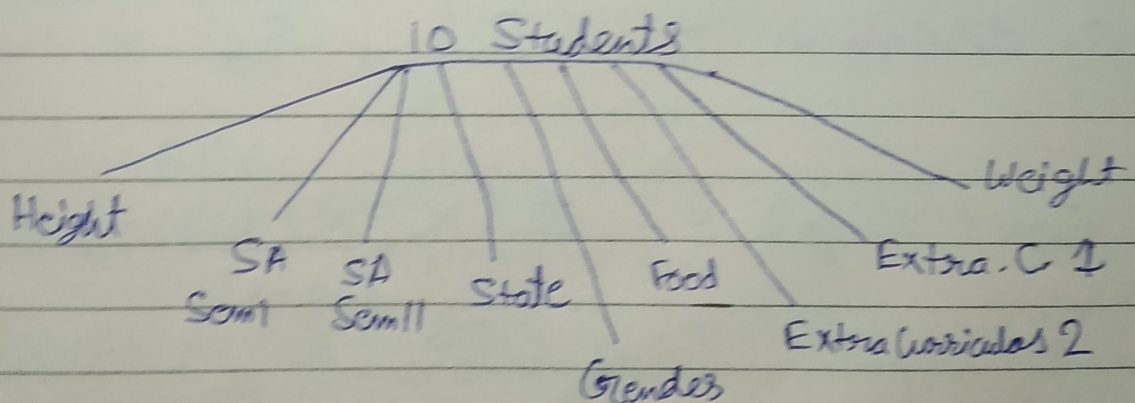
(Performance)  $P$ ,  $\frac{T}{(Task)}$ ,  $E$  (Experience)

Machine learning is the study of algorithm that can perform some task & improve performance by its experience.

Data Collection  $\rightarrow$  Data Labeling  $\rightarrow$  Feature instruction

Algorithm  $\leftarrow$  Data Analysis  $\leftarrow$

\* Mean, Variance, Covariance & standard deviation



Row is Pattern & Column is feature.

## Correlation

↳ Positive Correlation    ↳ Negative Correlation

↳ Zero Correlation.

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{Y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)} \quad \rightarrow \text{Bessel Correction}$$

$$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{(n-1)}$$

S. deviation -  $S_x$  ,  $S_y$

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$$

| x | y  |
|---|----|
| 2 | 20 |
| 3 | 60 |
| 6 | 80 |

| x | y   |
|---|-----|
| 2 | 100 |
| 4 | 80  |
| 6 | 60  |
| 8 | 30  |

| x | y |
|---|---|
| 1 | 4 |
| 2 | 5 |
| 3 | 4 |

Means

$$\bar{X}_1 = \frac{11}{3}$$

$$\bar{Y}_1 = \frac{160}{3}$$

~~Cov.~~

$$\bar{X}_2 = \frac{20}{4}, \frac{270}{4}$$

$$\Rightarrow 5 \quad \Rightarrow 67.5$$

$$\bar{X} = 2$$

$$\bar{Y} = 13/3$$

$$\begin{aligned} \text{Cov.} = & \frac{(2 - 11/3)(20 - 160/3) + (3 - 11/3)(60 - 160/3) + (6 - 11/3)(80 - 160/3)}{2} \\ & \Rightarrow 56.666 \end{aligned}$$

$$\frac{500}{9} + \left(\frac{-40}{9}\right) + \frac{560}{9}$$

$$\Rightarrow 56.666$$

$$\begin{aligned} \text{Cov} = & \frac{(2 - 5)(100 - 67.5) + (4 - 5)(80 - 67.5) + (6 - 5)(60 - 67.5) + (8 - 5)(30 - 67.5)}{3} \\ & \Rightarrow -76.667 \end{aligned}$$

$$\Rightarrow -76.667$$

$$\text{Cov}(x, y) = \frac{(1 - 2)(4 - 13/3) + (4 - 13/3)}{2}$$

$$\Rightarrow 0$$

| $x$ | $y$ |
|-----|-----|
| 101 | 100 |
| 81  | 80  |
| 61  | 60  |
| 51  | 50  |
| 41  | 40  |
| 21  | 20  |
| 1   | 0   |
| 51  | 50  |

| $x$ | $y$ |
|-----|-----|
| 54  | 53  |
| 53  | 52  |
| 52  | 51  |
| 51  | 50  |
| 50  | 49  |
| 49  | 48  |
| 48  | 47  |
| 51  | 50  |

$$\text{Cov}(x, y) = 1166.67$$

$$r_{xy} = \frac{\text{Cov}(x, y)}{S_x S_y}$$

| # | $x$ | $y$ |
|---|-----|-----|
| 2 | 6   | 6   |
| 3 | 6   | 6   |
| 8 | 9   | 9   |

Pearson Correlation Coefficient.

$$X = [-3, -2, -1, 0, 1, 2, 3]$$

$$y = x^2$$

$$\bar{x} = 0$$

- 1) What is meaning strong +ve & -ve Correlat.
- 2) Justify the reasons that Pearson's Correlation is used for linear only.

$\Rightarrow$  Correlation is the study of association.

$\Rightarrow$  Regression - is to predict dependent variable using independent variable.

- Polynomial Curve Fitting Problem.

# Advertisement (in lakh): 11, 13, 14, 16, 16, 15, 15  
14, 13, 13

Sales Volume (in lakh): 50, 50, 55, 60, 65, 65, 65  
60, 60, 50

$$\text{Mean (Ads)} = \frac{140}{10} = 14$$

$$\text{Mean (Sales)} = \frac{580}{10} = 58$$

$$\Rightarrow \text{Cov}(x, y) = \frac{(-3)(-8) + (-1)(-8) + 0(3) + 2(2) + 2(7) + 1(7) + 1(7) + 0(2) + (-1)(2) + (-1)(-8)}{(10-1)}$$

$$\Rightarrow \frac{24 + 8 + 4 + 14 + 7 + 7 + 7 - 2 + 8}{(10-1)}$$

$$\Rightarrow \frac{70}{9} = 7.77.$$

$$S_x = \sqrt{\frac{9 + 1 + 0 + 4 + 4 + 1 + 1 + 0 + 1 + 1}{10-1}}$$

$$\Rightarrow \sqrt{\frac{22}{10-1}} \Rightarrow \sqrt{\frac{22}{9}}$$

$$S_y^2 = \frac{64 + 64 + 4 + 49 + 49 + 49 + 4 + 64 + 9 + 4}{9}$$

$$\Rightarrow \frac{397}{9} = 40$$

$$S_y = \sqrt{40}$$

## Polynomial Curve Fitting -

available  $\langle x_i, y_i \rangle$

Predicting  $y$  for unknown  $x$

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

~~$a_0 + a_1$~~

$a_0, a_1, a_2, \dots, a_n$

Supervised learning - Finding good prediction using labelled data

2 problem types

Regression  
( $y$  is continuous)

Classification  
(If  $y$  is discrete)

## Linear Regression

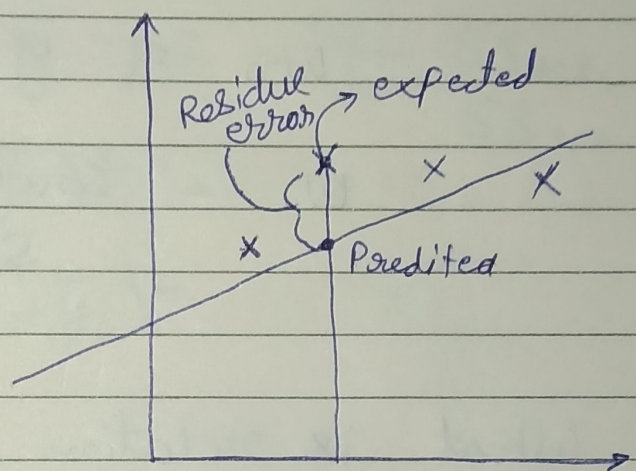
$$y = w_0 + w_1 x$$

| $x$ | $y$ |
|-----|-----|
| 2   | 0.8 |
| 3   | 0.9 |
| 3   | 0.8 |

$$\epsilon_i = y_i - \hat{y}_i$$

Residual  
error

RSS - Residual Square  
Sum



hypothesis  $\hat{y} = h(x) = w_0 + w_1 x$

Residual :  $\epsilon_i = y_i - \hat{y}_i \Rightarrow y_i = \hat{y}_i + \epsilon$   
 $\Rightarrow w_0 + w_1 x + \epsilon$

L.S.E =  $\sum_{i=1}^m (y_i - \hat{y}_i)^2$

Least Square Error

↪ Sum Square Errors.

$$\Rightarrow J(w_0, w_1) = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

↪ Sometimes

$$\frac{1}{2} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

## Ordinary linear regression.

$$\frac{\partial J(w)}{\partial w_0} = 0$$

$$\frac{\partial J(w)}{\partial w_1} = 0$$

$$w_0 = \bar{y} - w_1 \bar{x}$$

$$w_1 = \frac{\text{cov}(x, y)}{S_x^2} = r \cdot \frac{S_y}{S_x}$$

Solve

# What is relation b/w slope & correlation

## ② Multiple linear regression.

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_n x_{in} + \epsilon$$

hypothesis

$$h(x_i) = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_n x_{in}$$

$$x_i = [1, x_{i1}, x_{i2}, x_{i3}, \dots, x_{in}] \quad \hat{y} = \sum_{j=0}^n w_j x_{ij}$$

$$Y = XW + \epsilon$$

$$Y \Rightarrow m \times 1$$

$$X \rightarrow m \times (n+1)$$

$$W \rightarrow (n+1) \times 1$$

$$\epsilon \rightarrow m \times 1$$

$$W = (X^T X)^{-1} \cdot X^T Y$$

## ► Coefficient of determination.

It is the statistical measure that is used in linear regression that indicate how well the regression model explains the variability of dependent variable.

$$R^2 = \frac{1 - SS_{\text{res.}}}{SS_{\text{Tot}}}$$

$$SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2$$

$$SS_{\text{tot}} = \sum (y_i - \bar{y}_i)^2$$

$$R^2 = 1$$

Best fit

$$R^2 = 0$$

Worst fit

## # Estimate y using x by OLS.

# Why we normalize R<sup>2</sup>  
by  $S_x$  &  $S_y$  & not  
by  $S_x^2$  &  $S_y^2$

| X | Y   |
|---|-----|
| 1 | 1.2 |
| 2 | 1.8 |
| 3 | 2.6 |
| 4 | 3.2 |
| 5 | 3.8 |

OLS: Ordinary least square Error.  $x = 6$   
 $y = ?$

## Ordinary linear Regression -

$$\hat{y} = w_0 + w_1 x$$

↪ predicted

$$\hat{y}_i = w_0 + w_1 x_i$$

$$\text{residue} = \epsilon_i = y_i - \hat{y}_i \Rightarrow y_i - w_0 - w_1 x_i$$

$$(J) \text{ L.S.E} = \sum_{i=1}^n (y_i - w_0 - w_1 x_i)^2$$

$$\frac{\partial J}{\partial w_0} = -2 \sum_{i=1}^n (y_i - w_0 - w_1 x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n (y_i - w_0 - w_1 x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n y_i - n w_0 - w_1 \sum_{i=1}^n x_i = 0$$

$$\Rightarrow \sum y_i = n w_0 + w_1 \sum x_i$$

(Divide by n)  $\Rightarrow \bar{y} = \bar{a} + b \bar{x}$

~~$\bar{y} = a + b \bar{x}$~~

$$\frac{\sum y_i}{n} = w_0 + w_1 \frac{\sum x_i}{n}$$

$$\Rightarrow \bar{y} = w_0 + w_1 \bar{x}$$

$$w_0 = \bar{y} - w_1 \bar{x} \quad \text{--- (1)}$$

$$\frac{\partial J}{\partial w_1} = -2 \sum_{i=1}^n x_i (y_i - w_0 - w_1 x_i) = 0$$

$$\sum x_i (y_i - w_0 - w_1 x_i) = 0$$

$$\Rightarrow \sum x_i y_i - w_0 \sum x_i - w_1 \sum x_i^2 = 0$$

$$\Rightarrow \sum x_i y_i = w_0 \sum x_i + w_1 \sum x_i^2 \quad \text{--- (2)}$$

from (1)

$$\sum x_i y_i = \bar{y} \sum x_i - w_1 \bar{x} \sum x_i + w_1 \sum x_i^2$$

$$\sum x_i y_i = n \bar{x} \bar{y} - w_1 n \bar{x}^2 + w_1 \sum x_i^2$$

$$\sum x_i y_i - n \bar{x} \bar{y} = w_1 (\sum x_i^2 - n \bar{x}^2)$$

$$\Rightarrow w_1 = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2}$$

$$w_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\Rightarrow \frac{\text{Cov}(x, y)}{S_x^2}$$

# Estimate  $y$  using  $x$

$$x = 1, 2, 3, 4, 5$$

$$y = 1.2, 1.8, 2.6, 3.2, 3.8$$

let  $\hat{y} = a + bx$

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\bar{y} = \frac{1.2 + 1.8 + 2.6 + 3.2 + 3.8}{5} = 2.52$$

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\Rightarrow \frac{(-2)(-1.32) + (-1)(-0.72) + 0(0.08) + (1)(0.68) + 2(1.28)}{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}$$

$$\Rightarrow \frac{2.64 + 0.72 + 0 + 0.68 + 2.56}{4 + 1 + 0 + 1 + 4}$$

$$\Rightarrow \frac{6.60}{10} = 0.66$$

$$\bar{a} = \bar{y} - 0.66 \bar{x}$$

$$\Rightarrow 2.52 - 0.66(3)$$

$$\Rightarrow 2.52 - 1.98 = 0.54$$

$$\hat{y} = 0.54 + 0.66x$$

$$x = 6, \quad \hat{y} = 0.54 + 0.66 \times 6$$

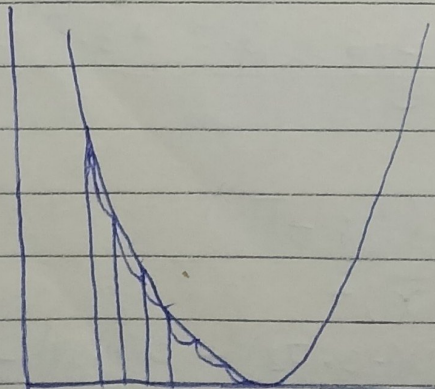
$$\Rightarrow 4.5$$

### ► Iterative Optimization

$$J(w) = \frac{1}{2m} \sum (y_i - \hat{y}_i)^2$$

$$\text{MSE}$$

$$w = [w_0, w_1]$$



MLR

$$Y = XW + \varepsilon$$

$$\text{Model: } \hat{y} = h(X) = XW$$

$$X_i = [1, x_{i1}, x_{i2}, x_{i3}, \dots, x_{in}]$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \dots & x_{1n} \\ 1 & x_{21} & - & - & - & - \\ 1 & x_{31} & - & - & - & - \\ \vdots & \vdots & - & - & - & - \\ 1 & x_{m1} & - & - & - & - \end{bmatrix}$$

$$\hat{y}_i = \sum_{j=0}^n w_j x_{ij} \quad | \quad x_{i0} = 1$$

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$w \rightarrow$  estimate / by minimizing  $J(w)$

$$w_j^{iter+1} = w_j^{iter} - \alpha \frac{\partial J(w)}{\partial w_j}$$

H.W.

$$w_j^{iter+1} = w_j^{iter} - \alpha \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij} \quad \text{in terms of } x_{ij}, y_i$$

## Types of Gradient descent.

### 1) Batch gradient descent

1 batch - whole training sample

### 2) Stochastic gradient descent.

$$W_j^{its+1} = W_j^{its} - \alpha (\hat{y}_i - y_i) x_{ij}$$

### 3) MiniBatch gradient descent.

$$W_j^{its+1} = W_j^{its} - \alpha \frac{1}{K} \sum_{i=1}^K (\hat{y}_i - y_i) x_{ij}$$

$$|J^{its}(w) - J^{its-1}(w)| > \rho$$

+ number of epochs  $\leq 100$

H.W

Incremental gradient descent.

What is limitation of SGD

## → Grid Search For Hyperparameters

- Set of Hyperparameters
- Train model using training set
- Validate using Validation set.
- Retain Best model
- Test with Testing

→ How to test over-fitting during training itself.

→ Too many features & less samples

## ② Feature Scaling -

Feature 1 : 1 - 5

Feature 2 : 10 - 10,000

Min-Max Scaling

$$X' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardization

$$X' = \frac{x - \mu}{\sigma}$$

Absolute Max

$$X' = \frac{x}{\max(|x|)}$$

# Design a linear regression model using batch GD Approach by considering following dataset & use this model to predict the output for test sample  $[3, 12]$

Sol

|  | $w_0$ | $w_1$ | $w_2$ | $x_1$ | $x_2$ | $y$ |
|--|-------|-------|-------|-------|-------|-----|
|  | 1     | 1     | 1     | 1     | 2     | 6   |
|  |       |       |       | 2     | 10    | 24  |

Dataset

$$w_j^{its+1} = w_j^{its} - \alpha \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij}$$

$$\alpha = 0.1$$

# training Step = 2

$$\hat{y}_i = w_0 + w_1 x_{i1} + w_2 x_{i2}$$

All parameters are initialised as 1

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 10 \end{bmatrix} \quad Y_i = \begin{bmatrix} 6 \\ 24 \end{bmatrix}$$

$$\hat{y}_1 = \begin{bmatrix} 4 \\ 13 \end{bmatrix}, \quad (\hat{y}_i - y_i) = \begin{bmatrix} -2 \\ -11 \end{bmatrix}$$

$$\hat{y}_2 = 1 + (1)(2) + (1)(10) = 13, \quad \text{error} = -11$$

$$w_0' = 1 - \frac{0.1}{2} [(-2)(1) + (-11)(1)]$$

$$\Rightarrow 1 + 0.65 = 1.65$$

$$w_1' = 1 - \frac{0.1}{2} [(-2)(1) + (-11)(2)]$$

$$\Rightarrow 1 + \frac{0.1}{2} (24) \Rightarrow 1 + 1.2 = 2.2$$

$$w_2' = 1 - \frac{0.1}{2} [(-2)(2) + (-11)(10)]$$

$$\Rightarrow 1 - \frac{0.1}{2} (-110) \Rightarrow 1 + 0.1 \times 55 = 6.7$$

$$w_0'' = \hat{y}_1 = -1.65$$

$$\hat{y}_1 = 1.65 + (2.2)(1) + (6.7)(2) \\ \Rightarrow 17.25$$

$$\hat{y}_1 - y_1 = 17.25 - 6 = 11.25$$

$$\hat{y}_2 = 1.65 + (2.2)(2) + 6.7(10) \Rightarrow 73.05$$

error  $(\hat{y}_2 - y_2) = 73.05 - 24 = 49.05$

$$w_0'' = w_0' - \frac{\alpha}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{i0}$$

$$\Rightarrow 1.65 - \frac{0.1}{2} [11.25(1) + (49.05)(1)]$$

$$\Rightarrow 1.65 - 0.05(60.3)$$

$$\Rightarrow -1.365$$

$$w_1'' = w_1' - \frac{\alpha}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{i1}$$

$$\Rightarrow 2.2 - \frac{0.1}{2} [(11.25)(1) + (49.05)(2)]$$

$$\Rightarrow 2.2 - 0.05[11.25 + 98.10]$$

$$\Rightarrow 2.2 - 0.05[109.35]$$

$$\Rightarrow 2.2 - 0.05(109.35)$$

$$\Rightarrow -3.2675$$

$$w_2'' = 6.7 - 0.05 [11.25(2) + (49.05)(10)]$$

$$\Rightarrow 6.7 - 0.05 [22.5 + 490.5]$$

$$\Rightarrow 6.7 - 25.65 \Rightarrow -18.95$$

$$y = -1.365 + (-3.2675)(3) + (-18.95)(12) = -238.5625$$

## Logistic Regression (Used for binary class.)

$$h(x_i) = \hat{y}_i \geq 0.5 \quad 1$$

$$\hat{y}_i < 0.5 \quad 0$$

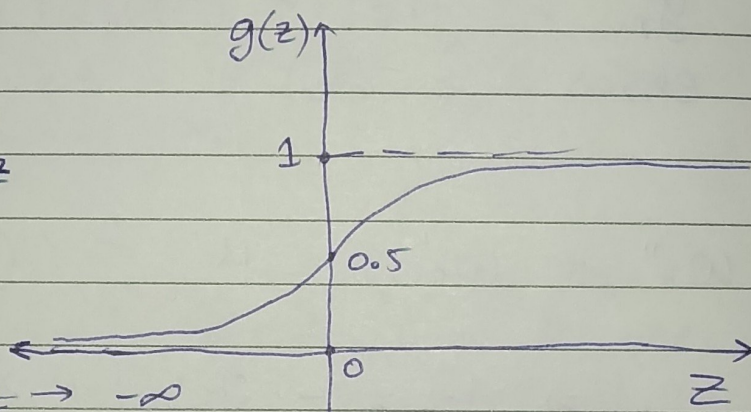
### Logistic function

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g(0) = 0.5$$

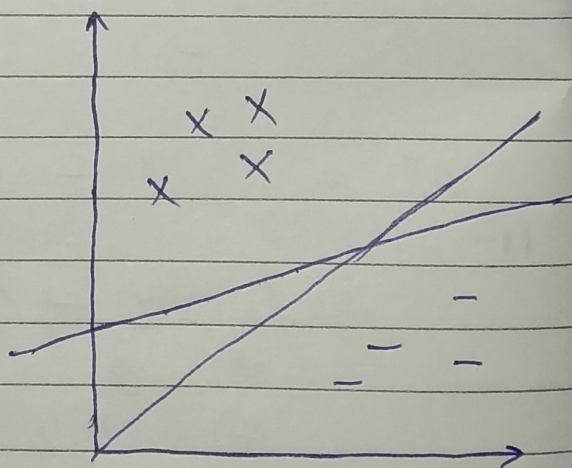
$$g(z) \rightarrow 0 \quad z \rightarrow -\infty$$

$$g(z) \rightarrow 1 \quad z \rightarrow \infty$$



$$w_1 x_1 + w_2 x_2 + w_0 \geq 0$$

$$w_1 x_1 + w_2 x_2 + w_0 < 0$$



$$\sum w_j x_{ij} \geq 0$$

$$\sum w_j x_{ij} < 0$$

$$\hat{y}_i = h(x) = g\left(\sum w_j x_{ij}\right)$$

# Considering MSE cost func. if we apply batch GD what will be the formula for updation.

$$MSE = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$W_j^{it_{g+1}} = W_j^{it_g} - \alpha \frac{\partial}{\partial W_j} J(w)$$

| $x_1$ | $x_2$ | Clean label |   |   |
|-------|-------|-------------|---|---|
| 2     | 0.6   | Spam        | 1 | 1 |
| 6     | 3     | Not Spam    | 2 | 0 |
| 1     | 2     | Spam        | 1 | 1 |

$$W_j^{it_{g+1}} = W_j^{it_g} + \alpha (y_i - \hat{y}_i) x_{ij} \hat{y}_i (1 - \hat{y}_i) \quad \text{SGD}$$

$$\Rightarrow W_j^{it_g} + \frac{\alpha}{m} \sum (y_i - \hat{y}_i) x_{ij} \hat{y}_i (1 - \hat{y}_i) \quad \text{BGD}$$

# What is the function used for calculating loss in multiple

log-loss  
cost function

$$J(w) = \frac{1}{m} \sum_{i=1}^m \text{cost}(\hat{y}_i, y_i)$$

$$\text{cost}(\hat{y}_i, y_i) = \begin{cases} -\log(\hat{y}_i) & \text{if } y_i = 1 \\ -\log(1 - \hat{y}_i) & \text{if } y_i = 0 \end{cases}$$

$$J(w) = \frac{1}{m} \sum_{i=1}^m -y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)$$

↳ for binary.

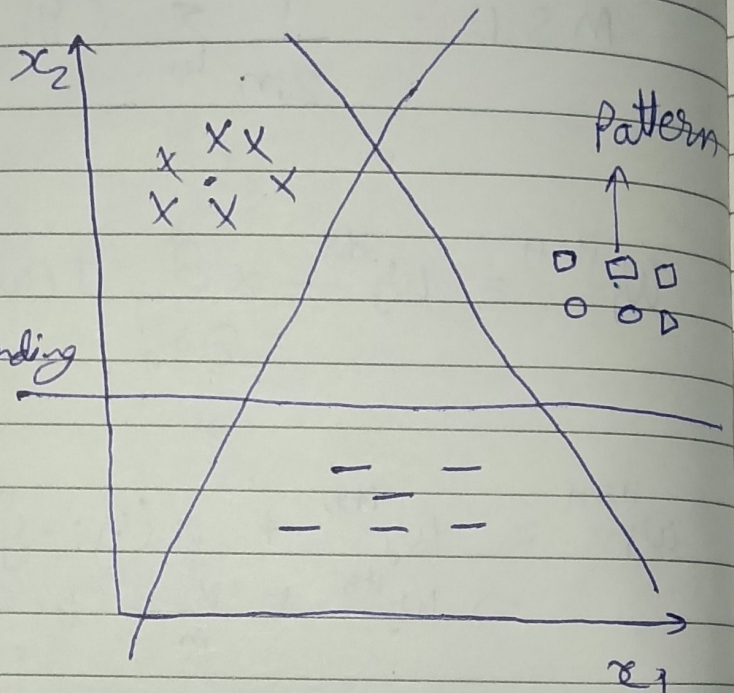
# # Difference b/w Loss function / Cost function

- 2 Feature 3 classes.

For Training -

For the first one.  
train a logistic  
regression model

$h^k(x_i)$  corresponding  
to class  $k$  to  
predict probability  
that  $y_i = k$   
for a given  $x_i$



For testing -  
 $x_{\text{test}}$

$h^1(x_{\text{test}})$   
 $h^2(x_{\text{test}})$   
 $\vdots$   
 $h^c(x_{\text{test}})$

Choose  $k$  for

$\max \{ h^k(\text{test}) \}$

## K fold validation -

- 1) Shuffle the data
- 2) Divide sample in k size
- 3) Keep ① set as validation & train with others.
- 4) Get train & test value.

## Confusion Matrix -

|             |   | (Actual) |    |
|-------------|---|----------|----|
|             |   | +        | -  |
| (Predicted) | + | TP       | FP |
|             | - | FN       | TN |

$$\text{Accuracy} =$$

$$\frac{TP + TN}{TP + FP + FN + TN}$$

$$\text{Acc}_{(+)} = \frac{TP}{TP + FN}$$

$$\text{Acc}_{(-)} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

#

Actual

|          |      | Cat | Fish | Dog |
|----------|------|-----|------|-----|
| Predict. | Cat  | 4   | 6    | 3   |
|          | Fish | 1   | 2    | 0   |
|          | Dog  | 1   | 2    | 6   |

$$\text{Precision}_{\text{cat}} = \frac{4}{4+6+3}$$

$$\text{Recall}_{\text{cat}} = \frac{4}{4+1+1}$$

$$\text{Fish} = \frac{2}{1+2+3}$$

$$\text{Fish} = \frac{2}{6+2+2}$$

$$\text{Dog} = \frac{6}{1+2+6}$$

$$\text{Dog} = \frac{6}{3+0+6}$$

| #     | $x_1$ | $x_2$ | $y$ |
|-------|-------|-------|-----|
| $P_1$ | 1     | 2     | 1   |
| $P_2$ | 2     | 1     | 0   |

$$w_0 = 0 \quad w_1 = 0.1$$

$$w_2 = -0.2 \quad X = 0.1$$

$$\text{epoch} = 2$$

Sol

$$\sum w_i x_{ij} h(P_1) = 0 + 0.1(1) + (-0.2)2$$

$$\Rightarrow 0.1 - 0.4 \Rightarrow -0.3$$

$$\sum w_i x_{ij} h(P_2) = 0 + 0.1(2) + (-0.2)1$$

$$\Rightarrow 0.2 - 0.2 = 0$$

$$h(P_1) = \frac{1}{1 + e^{0.3}} = 0.4255, \quad h(P_2) = \frac{1}{1 + 1} = 0.5$$